Mon 24 March 2025

Nonlinear Equations

Ch 5 in Heath.

Problem: Determine roots of a general equation of the form

$$f(x) = 0.$$

That is, find x^* such that $f(x^*) = 0$.

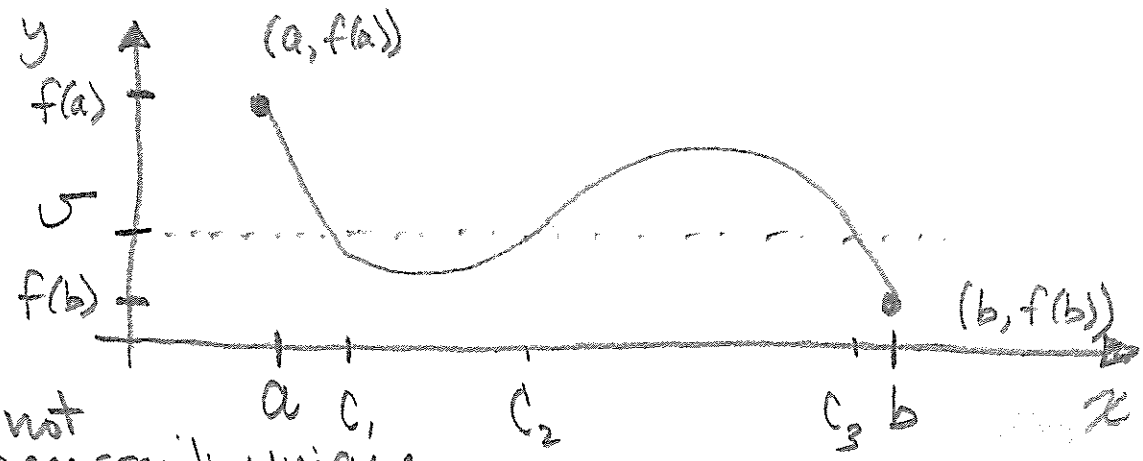
We'll stick with looking at a single equation $f: \mathbb{R} \to \mathbb{R}$ (since methods for $g: \mathbb{R}^n \to \mathbb{R}^n$ require use multivariate calculus)

Before we put effort in to computing roots, we should ask when/how can we know that a root exists. Fortunately, a theorem from first year calculus assists.

Intermediate Value Theorem (IVT)

If $f \in C[a,b]$, and v is any number between $f(a)$ and $f(b)$, then there exists a number $c \in [a,b]$ for which $f(c) = v$.

pictorally:



note: c is not necessarily unique.

We can apply the IVT to get an algorithm for finding roots.

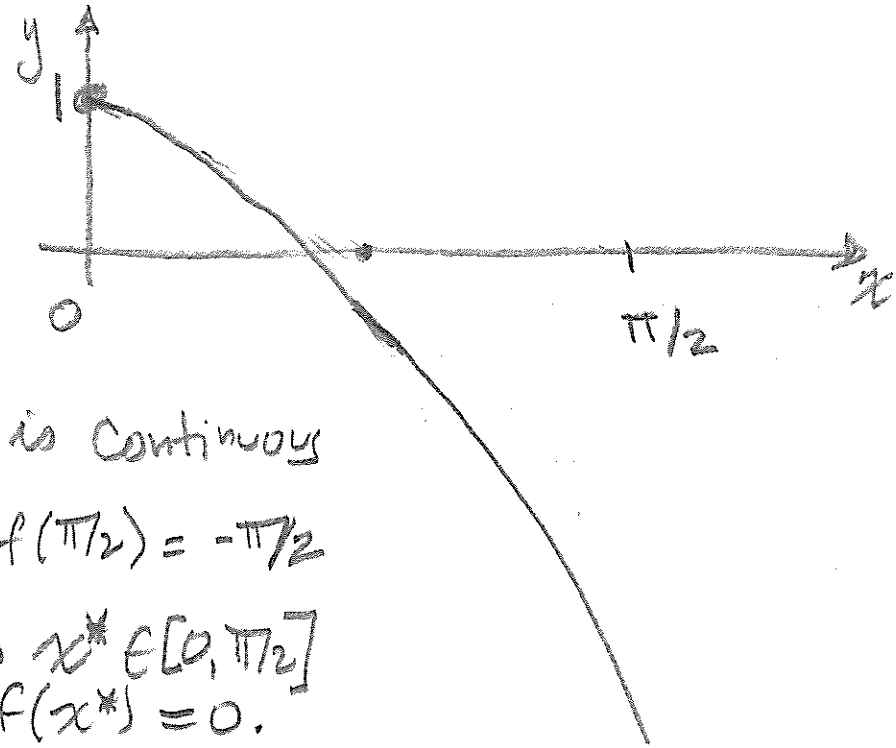
If f is continuous, and we have two values of x

$$x = L \quad \text{and} \quad x = R \qquad (L < R)$$

that are such that $f(L)$ and $f(R)$ have opposite sign, then, since 0 must be between $f(L)$ and $f(R)$, there must be an x^* between L and R with $f(x^*) = 0$.

<u>example</u> Consider $f(x) = \cos(x) - x$

Pictorially:



Clearly $f(x)$ is continuous

and $f(0) = 1$, $f(\pi/2) = -\pi/2$

so there is an $x^* \in [0, \pi/2]$ such that $f(x^*) = 0$.

Consider the mid point of this interval:

$$m_1 = 0 + \frac{\pi/2 - 0}{2}$$
$$= \pi/4.$$

If we were to approximate x^* by m_1, then the absolute error in the approximation is

$$e_1 = |m_1 - x^*|$$
$$\leq \tfrac{1}{2}\,|\pi/2 - 0| \quad \text{(the interval width)}$$
$$= \pi/4$$
$$\doteq 0.785.$$

We would likely want a more accurate approximation.

Let's compute: $f(m_1) = f(\pi/4)$
$$= \cos(\pi/4) - \pi/4$$
$$\doteq 0.7071 - 0.7854$$
$$= -0.07829$$

$\therefore$ by the IVT, there must be a root in $[0, \pi/4]$.

Then $m_2 = 0 + \frac{\pi/4 - 0}{2}$ and $e_2 = |\pi/8 - x^*|$
$$= \pi/8 \qquad\qquad \leq \tfrac{1}{2}\,|\pi/4 - 0|$$
$$= \pi/8$$
$$\doteq 0.393$$

$f(m_2) = \cos \pi/8 - \pi/8$
$$\doteq 0.5312$$

and therefore there must be a root in $[\pi/8, \pi/4]$.

We can repeat this process and summarize in a table.

bracket

iteration i	L_i	R_i	m_i	$f(L_i)$	$f(R_i)$	$f(m_i)$
1	0	$\pi/2$	$\pi/4$	+	−	−0.078
2	0	$\pi/4$	$\pi/8$	+	−	+0.531
3	$\pi/8$	$\pi/4$	$3\pi/16$	+	−	+0.242
4	$3\pi/16$	$\pi/4$	$7\pi/16$	+	−	+0.086

etc.

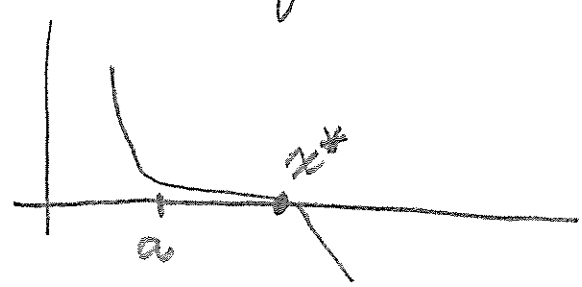
At each iteration, the error in the approximation of x^* by m_i is $e_i \leq \frac{1}{2}|R_i - L_i|$

$\boxed{\text{Issue:}}$ When should we stop the iteration?

A^1: When $f(m)$ is "small enough"

But this approach can be fooled if $f(x)$ is flat near the root.

$f(a)$ is small
but $|a - x^*|$ is large



A^2: Stop when bound on error is sufficiently small

e.g. $\frac{1}{2}|R - L| \leq$ ABSTOL

then take $x^* = L_i + \frac{1}{2}|R_i - L_i|$

(Other thoughts: • use Relative Error instead of Abs Error
• put in max value on # iterations

This algorithm is called the Bisection Method.:

given: continuous $f(x)$, L, R s.t. $f(L)\cdot f(R) < 0$, $L < R$

while $\frac{1}{2}(R-L) > \text{ABSTOL}$:

$m = L + (R-L)/2$

if $\text{sign}(f(L)) == \text{sign}(f(m))$:

$L = M$

else

$R = M$

$x^* = L + (R-L)/2$.

Reasonable Question: How quickly does the method converge?

We have
$$e_1 = |m_1 - x^*| \le \tfrac{1}{2}(R_1 - L_1)$$
$$e_2 = |m_2 - x^*| \le \tfrac{1}{2}(R_2 - L_2) = \tfrac{1}{4}(R_1 - L_1)$$
$$e_3 = |m_3 - x^*| \le \tfrac{1}{2}(R_3 - L_3) = \tfrac{1}{8}(R_1 - L_1)$$

After finding n intervals, we have
$$e_n = |m_n - x^*| \le \frac{1}{2^n}|R_1 - L_1|$$

<u>example</u>: Suppose we need to find a root of
$$f(x) = x - 0.2\sin(x) - 0.5$$
with a final error bound of $|\bar{x} - x^*| \le 5\times 10^{-7}$

I will cover on Wed.

Computing $f(0) = -0.5$ and $f(1) = +0.3317$, we know that there must be a root in $[0,1]$. How many iterations of the bisection method may be required to meet the error tolerance?

<u>Solution</u>: We need to find an n such that

$$e_n \leq \frac{1}{2^n}(1-0) \leq 5\times10^{-7}$$

$$\therefore\ 2^n \geq 0.2\times10^{7}$$

$$n \geq \log(0.2\times10^{7})/\log(2)$$

$$\doteq 20.93$$

$\therefore$ we would need 21 iterations of the bisection method.

We can confirm this by reviewing the table of computational results in the $\boxed{\text{handout.}}$

The plot of $f(x)$ looks ~linear, which suggests we are doing a lot of work to find the root!

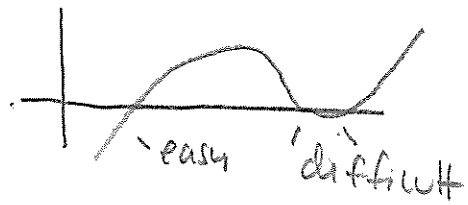
The Bisection Method

<u>good features</u>

- given an initial bracket of the root, method guaranteed to converge to root. (will see not all methods give this)
- only requires knowledge of $f(x)$ (will see some methods require derivative)
- can predict in advance how many iterations are req'd to meet error tolerance
- easy to implement

<u>bad features</u>

- requires starting with a bracket of root, which may be difficult for some roots.

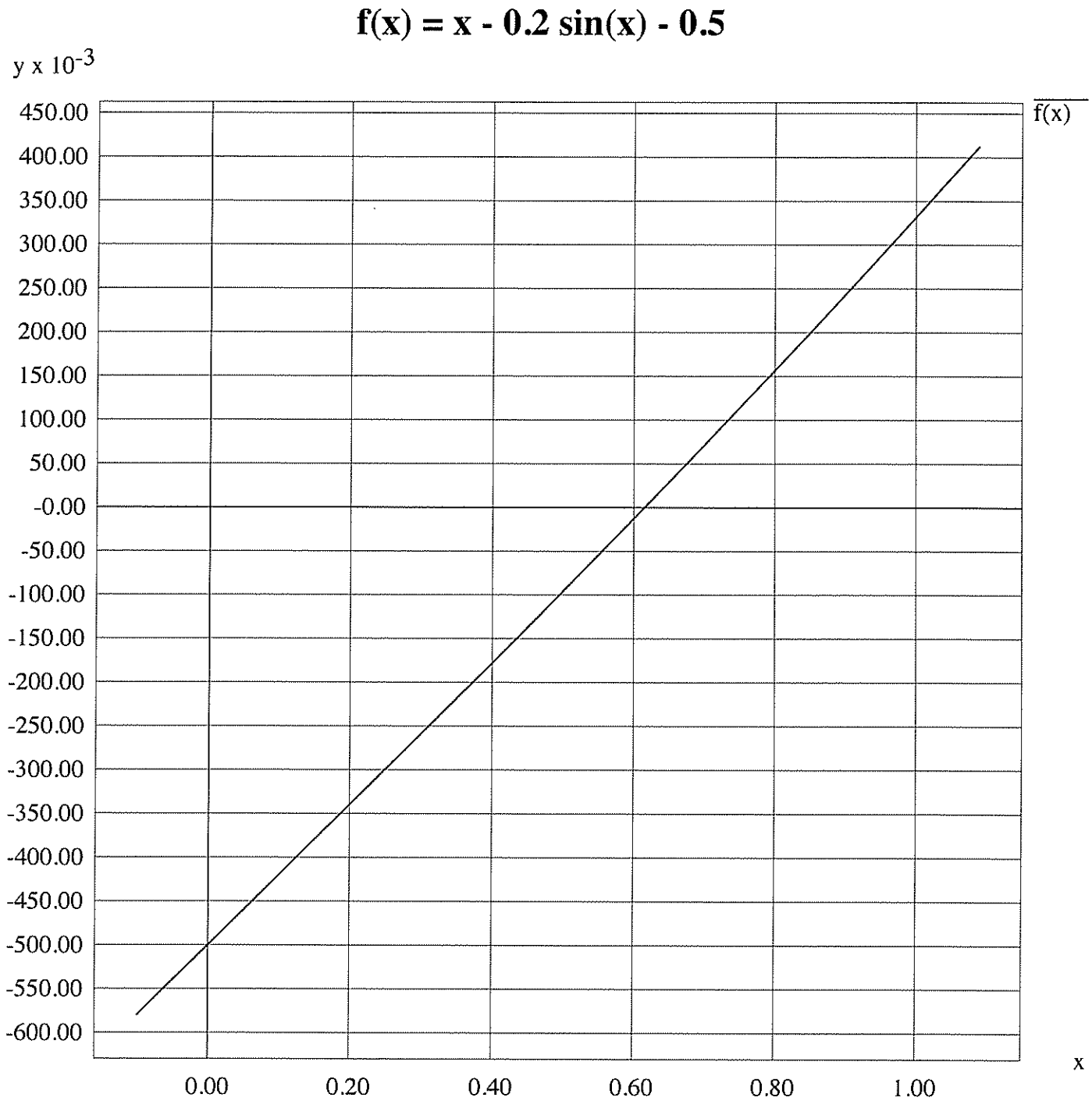


- converges <u>slowly</u> and not <u>uniformly</u> to root. (next iteration may be further away, $f(m)$ bounces around!)
- does not use all available info. (only the sign of $f(\,)$) (1 bit of info)

Given the "bad" features, we'll look at other methods!

Bisection applied to $f(x) = x - 0.2\sin(x) - 0.5$

step number	L	$f(L)$	R	$f(R)$	M	$f(M)$	max error
1	0.000000E+00	-	1.000000E+00	+	5.000000E-01	-9.588511E-02	5.0E-01
2	5.000000E-01	-	1.000000E+00	+	7.500000E-01	1.136722E-01	2.5E-01
3	5.000000E-01	-	7.500000E-01	+	6.250000E-01	7.980545E-03	1.2E-01
4	5.000000E-01	-	6.250000E-01	+	5.625000E-01	-4.416053E-02	6.2E-02
5	5.625000E-01	-	6.250000E-01	+	5.937500E-01	-1.814463E-02	3.1E-02
6	5.937500E-01	-	6.250000E-01	+	6.093750E-01	-5.096014E-03	1.6E-02
7	6.093750E-01	-	6.250000E-01	+	6.171875E-01	1.438734E-03	7.8E-03
8	6.093750E-01	-	6.171875E-01	+	6.132812E-01	-1.829518E-03	3.9E-03
9	6.132812E-01	-	6.171875E-01	+	6.152344E-01	-1.956125E-04	2.0E-03
10	6.152344E-01	-	6.171875E-01	+	6.162109E-01	6.215054E-04	9.8E-04
11	6.152344E-01	-	6.162109E-01	+	6.157227E-01	2.129326E-04	4.9E-04
12	6.152344E-01	-	6.157227E-01	+	6.154785E-01	8.656611E-06	2.4E-04
13	6.152344E-01	-	6.154785E-01	+	6.153564E-01	-9.347883E-05	1.2E-04
14	6.153564E-01	-	6.154785E-01	+	6.154175E-01	-4.241132E-05	6.1E-05
15	6.154175E-01	-	6.154785E-01	+	6.154480E-01	-1.687741E-05	3.1E-05
16	6.154480E-01	-	6.154785E-01	+	6.154633E-01	-4.110413E-06	1.5E-05
17	6.154633E-01	-	6.154785E-01	+	6.154709E-01	2.273095E-06	7.6E-06
18	6.154633E-01	-	6.154709E-01	+	6.154671E-01	-9.186596E-07	3.8E-06
19	6.154671E-01	-	6.154709E-01	+	6.154690E-01	6.772177E-07	1.9E-06
20	6.154671E-01	-	6.154690E-01	+	6.154680E-01	-1.207210E-07	9.5E-07
21	6.154680E-01	-	6.154690E-01	+	6.154685E-01	2.782483E-07	4.8E-07


f(x) = x - 0.2 sin(x) - 0.5
y x 10-3
450.00
400.00
350.00
300.00
250.00
200.00
150.00
100.00
50.00
-0.00
-50.00
-100.00
-150.00
-200.00
-250.00
-300.00
-350.00
-400.00
-450.00
-500.00
-550.00
-600.00
f(x)
0.00
0.20
0.40
0.60
0.80
1.00
x